Kai TANG

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Education

The University of Hong Kong (HKU) **Hong Kong SAR, China**
PhD in Robotics, Learning, and Control *09/2022–Present*
- Department of Electrical and Electronic Engineering
- PhD Postgraduate Scholarship (PGS)

Imperial College London (IC) **London, United Kingdom**
*MSc in Control and Optimisation (**Distinction**)* *10/2020–10/2021*
- Department of Electrical and Electronic Engineering
- Final Project (Thesis) published in *IFAC World Congress 2023*

South China University of Technology (SCUT) **Guangzhou, China**
BEng in Process Equipment & Control Engineering *09/2016–07/2020*
- School of Mechanical and Automotive Engineering
- **National Scholarship**, GPA & Weighted Average: 3.7/4.0 & 87.3/100
- **Excellent Final Project (Thesis)** of the University

Exchange & Working Experience

Centre for Transformative Garment Production **Hong Kong SAR, China**
Student Research Assistant *03/2022–10/2025*
- Collaborated with Tohoku University, Sendai, Japan on robotic automation for garment manufacturing
- Organized and presented my research outcome at 3 international workshops
- Contributed to spin-off company SewingDX Ltd. for commercializing robotic sewing technologies

McGill University **Montreal, Canada**
Winter Program in Artificial Intelligence *01/2020–02/2020*
- Achieved A (Merit) in the course of *Introduction to Machine Learning*
- Designed and implemented intelligent system based on Deep Reinforcement Learning

Xiamen Hongfa Electroacoustic Ltd **Xiamen, China**
Production Engineering Intern *07/2019–08/2019*
- Gained hands-on experience in industrial manufacturing processes and production line operations
- Assisted in control system design and embedded system development for automated production equipment

Honors and Awards

2021: **Distinction** for MSc Degree at IC (70%+ for both Coursework and Final Project (Thesis))

2019: **National Scholarship** (TOP 2%), **2017, 2018** University Scholarships

2020: **Excellent Graduation Project (Thesis)** of SCUT (TOP 5%)

2019: Honorable Mention, International Mathematical Contest in Modeling (MCM)

2017–2019: Merit Student & Outstanding Student Cadre (SCUT)

Research Projects

The University of Hong Kong
PhD Research Project *09/2022–Present*
"Robot learning and control for fabric manipulation and fixture-free automated sewing"
- Full stack system development using dual-arm manipulator (C++/Python/EtherCAT/Ethernet/lerobot)
- Time-scaling nonlinear modeling and control, dual-arm impedance control, and discrete event control
- Deep learning based multi-level perception, and imitation learning based policy learning
- Published 3 first-author papers (IEEE/ASME T-MECH, IEEE RA-L, 1 arXiv) and 1 International patent
- Deeply collaborated on 5 additional projects on hardware design, learning and control

Imperial College London

Research Project (MSc Thesis) *01/2021–09/2022*

"Decentralised model predictive control for autonomous intersection crossing"

- Designed a decentralised MPC framework for different types of vehicles to cross signal-free intersections
- Utilized nonlinear control, optimization, convexification and linearization techniques (MATLAB/Python)

South China University of Technology

Graduation Project (BS Thesis) *01/2020–06/2020*

"S22 microphone Injection mold design"

- Achieved Excellent Graduation Project (Thesis) (TOP 5% for whole university)
- Designed a novel injection mold based on CAD, CAE and CAM technologies (SolidWorks/Siemens UG).

National College Students' Innovation and Entrepreneurship Competition *05/2018–05/2019*

"The theory and systematic research on adaptive control of 3D printing"

- As the team leader achieved 1 Invention Patent and 4 Utility Model Patents
- Developed an adaptive slicing algorithm and layer optimization software for 3D printing
- Designed and optimized the mechanical structure and control system of the Coxy 3D printer

Publications & Patents

2025: **K. Tang**, D. Bhattacharya, H. Xu, F. Tokuda, N.C. Tien, K. Kosuge, "RTFF: Random-to-Target Fabric Flattening Policy using Dual-Arm Manipulator," *arXiv preprint*.

2025: **K. Tang**, X. Huang, A. Seino, F. Tokuda, A. Kobayashi, N.C. Tien, K. Kosuge, "Fixture-free automated sewing system using dual-arm manipulator and high-speed fabric edge detection," *IEEE Robotics and Automation Letters*.

2024: **K. Tang**, F. Tokuda, A. Seino, A. Kobayashi, N.C. Tien, K. Kosuge, "Time-scaling modeling and control of robotic sewing system," *IEEE/ASME Transactions on Mechatronics*.

2023: **K. Tang**, W. Wang, X. Pan, B. Chen, S.A. Evangelou, "Economic Potential for Hybrid Electric Vehicles in Urban Signal-free Intersections with Decentralized MPC," *IFAC World Congress*.

2025: K. Kosuge, **K. Tang**, A. Seino, F. Tokuda, A. Kobayashi, H. Pang NG, "Automated sewing system and controlling method thereof," *WO Patent*.

2025: E. Lo, X. Huang, **K. Tang**, A. Seino, F. Tokuda, K. Kosuge, "Fabric Flattening and Alignment System Using Real-Time Mesh-Based State Estimation and Visual Servoing," *IEEE International Conference on Mechatronics and Automation (ICMA)*, 2025. (**Best Conference Paper Award**)

2025: F. Tokuda, A. Seino, A. Kobayashi, **K. Tang**, K. Kosuge, "Transformer Driven Visual Servoing for Fabric Texture Matching Using Dual-Arm Manipulator," *IEEE Robotics and Automation Letters*.

2025: W. Dong, D. Bhattacharya, **K. Tang**, A. Kobayashi, F. Tokuda, A. Seino, K. Kosuge, "A Novel Fabric Alignment System for Sewing," *arXiv preprint*.

2025: A. Seino, A. Kobayashi, F. Tokuda, **K. Tang**, K. Kosuge, "Fabric Edge Folding along a Curved Line with Multiple Pleats," *arXiv preprint*.

Skills

Programming: Python, C++, PyTorch, MATLAB, lerobot, ROS2, Prompt Engineering

Robotics: EtherCAT, INTime, Servo Motor control, Force/Torque Sensor, Kinematics, Dynamics

Machine Learning: Deep Learning, Imitation Learning, Computer Vision, Simulation

Control Systems: Nonlinear Control, Optimization, Model Predictive Control, System Identification

Tools: SolidWorks, Siemens UG, AutoCAD, Git/GitHub, Linux, Docker, LaTeX, Adobe, Inkscape

Leadership & Activities

2017–2019: Class Monitor, School of Mechanical and Automotive Engineering, SCUT

2016–2019: Committee Member, School Youth Volunteers Guidance Center, SCUT

2016–2019: Leader, Human Resource Department, Students' Union, SCUT

Interests: Ball Sports, Traveling, Photography, and Music